

Master Interview Guide: Basketball Player Detection & Tracking

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1. Introduction: What is this project?

At its core, this project is a Computer Vision pipeline. It takes raw video footage of a basketball game and automatically identifies (detects) players and the basketball, follows them as they move (tracking), and generates analytical data about the game.

The problem it solves is manual labor. Analyzing basketball footage by hand to see where players spend time on the court takes hours. Commercial tools exist but are expensive and closed-source. This project is an automated, open-source alternative.

2. Core Technologies Used

YOLOv11m (You Only Look Once - version 11 medium)

- * What it is: A state-of-the-art Object Detection model. Object detection means the model draws "bounding boxes" around things it recognizes in an image.
- * Why it was chosen: YOLO is famous for its speed. In basketball, you need to process video quickly (real-time or near real-time). YOLO can run at 30 to 100 frames per second (FPS). Other models, like Faster R-CNN, might be slightly more accurate but only run at 5 to 10 FPS, which is too slow for video.
- * The "m" in YOLOv11m: Stands for "medium". YOLO models come in different sizes (nano, small, medium, large, extra-large). "Medium" is a sweet spot-it provides better accuracy than the smaller models without being too heavy or slow to run.

ByteTrack

- * What it is: A Multi-Object Tracking (MOT) algorithm. After YOLO detects the players in frame 1, frame 2, frame 3, etc., ByteTrack's job is to connect those detections. It assigns a unique ID (e.g., Player #4) to a person and keeps that ID attached to them as they move.
- * Why it was chosen: Basketball has a huge problem called "occlusion" (players blocking each other). When Player A steps in front of Player B, YOLO might momentarily lose confidence that Player B is there.
- * Older trackers (like SORT) throw away low-confidence detections and lose the player entirely.
- * ByteTrack is smarter. It uses a "two-stage matching pass." If it can't find a player using high-confidence detections, it looks at the low-confidence detections to re-find the partially hidden player. This keeps the player's ID consistent even when they are screened.

OpenCV (Open Source Computer Vision Library)

- * What it is: A massive library of programming functions mainly aimed at real-time computer vision.
- * How it's used: It's the engine for reading the video frames, drawing the bounding boxes, drawing the heatmaps, and saving the final processed video back to the computer.

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COCO Dataset (Common Objects in Context)

- * What it is: A massive dataset created by Microsoft containing over 330,000 images and 80 different object categories (including "person").
- * How it was used (Transfer Learning): Instead of training YOLOv11 from scratch (which takes days and huge amounts of data), this project used a YOLO model that was *already trained* on COCO. Because the COCO-trained model already knows what a "person" looks like, it only needed to be "fine-tuned" on the basketball dataset to learn the specific nuances of basketball players and the ball. This reduced training time to just 90 minutes.

3. Data Processing Pipeline

Machine learning models are only as good as the data they are trained on. Here is how the data was handled:

1. Merging: Three different datasets from Roboflow were combined to create a final dataset of about 3,600 images. This ensured the model learned from different courts, lighting, and angles.
2. Preprocessing: All images were resized to 640x640 pixels so the YOLO model received uniform input.
3. Quality Filtering & Edge Cases: Blurry or dark images were removed. Most importantly, the "Referee" class was dropped. Why? Referees standing near basketball posts were confusing the model, causing it to detect posts as people. Furthermore, an "aspect ratio filter" was used to remove any bounding boxes that were shaped like tall, skinny posts rather than humans.

4. Evaluation Metrics (The Math of Accuracy)

When training an object detection model, you must measure how well it is doing. Here are the key metrics used, and what they mean:

A. Precision

- * Concept: "When the model guesses 'Player', how often is it actually a Player?" It measures the quality of the positive predictions.
- * Formula: $\text{True Positives} / (\text{True Positives} + \text{False Positives})$
- * *Project Result: 95%+* (Very few "false alarms" like mistaking a post for a player).

B. Recall

- * Concept: "Out of all the actual Players on the court, how many did the model find?" It measures the quantity of true objects found.
- * Formula: $\text{True Positives} / (\text{True Positives} + \text{False Negatives})$
- * *Project Result: 94%+* (Very few missed players).

C. F1 Score

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- * Concept: A single number that balances Precision and Recall. If you have great precision but terrible recall, your F1 score will be bad.
- * Formula: $2 * ((\text{Precision} * \text{Recall}) / (\text{Precision} + \text{Recall}))$

D. mAP@50 (Mean Average Precision at 50% IoU)

- * Concept: The ultimate single metric for object detection.
- * IoU (Intersection over Union): Measures how much the model's bounding box overlaps with the *actual* (ground truth) bounding box.
- * @50: Means we consider a detection "Correct" if the overlap (IoU) is at least 50%.
- * Average Precision: The area under the Precision-Recall curve for a single class (e.g., just "Players").
- * Mean: The average of the Average Precisions across *all* classes (Players + Ball).
- * *Project Result: 97.2%* (This is an exceptionally high score, indicating the model is highly accurate at finding and tightly boxing players and the ball).

5. Final Analytics and Outcomes

The true value of this project isn't just drawing boxes; it's the data generated:

1. Tracked Video: A visual output where every player has a persistent ID and a trailing line showing their recent movement.
2. Heatmaps: A court overlay showing the exact areas where players spent the most time.
3. Zone Occupancy: Mathematical breakdowns of where play happened (e.g., 40% in the paint, 30% mid-range, 30% three-point zone).
4. Dashboards: Graphs showing track continuity, confidence levels over time, and player density.

Summary for the Interviewer: You built an automated pipeline using YOLOv11 for fast detection and ByteTrack for robust tracking through occlusion, achieving 97.2% mAP. You successfully navigated data cleanliness issues (the referee/post problem) and translated raw video pixels into actionable sports analytics.